Assignment 4: Maximum Concurrent Flow and Network Design

CS 422: Computer Networks, Fall 2026
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Due Date: Nov 06, 2026 @ 11:45 PM Eastern Time

Best Topology (100 points)

Consider a network design problem, where we have n nodes each with d incoming and d outgoing links. Each link is of capacity 1 unit. Given any traffic matrix, your goal is to design a network topology that maximizes the concurrent flow for the given traffic matrix. Throughout this assignment, "nodes" are sources and destinations, there are no additional intermediate switches.

- **Uniform Traffic Matrix (50 points)** Find the best topology that maximizes the concurrent flow for a uniform traffic matrix of size $n \times n$ defined as follows:

$$T = \begin{bmatrix} 0 & \frac{d}{n-1} & \cdots & \frac{d}{n-1} \\ \frac{d}{n-1} & 0 & \cdots & \frac{d}{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{d}{n-1} & \frac{d}{n-1} & \cdots & 0 \end{bmatrix}$$

  The answer must describe how to construct the topology (graph) either formally, or in words as a pseudocode. Explain why the chosen topology maximizes the concurrent flow for the given traffic matrix. You can assume that $n-1$ is divisible by d just for simplicity. You do not need to do a formal proof (although desirable here), but the grading depends on what arguments you use to justify your answer. You can also include a figure to illustrate the topology.

- **Arbitrary Traffic Matrix (50 points)** Find the best topology that maximizes the concurrent flow for any given traffic matrix of size $n \times n$ in a network with n nodes with d incoming and d outgoing links. Set $n = 8$ and $d = 4$. The traffic matrices are drawn from the following hose-model traffic set:

$$\mathcal{T}_{\text{hose}} = \left\{ T \in \mathbb{R}_{\geq 0}^{8\times 8} \;\middle|\; T_{ii} = 0,\ \forall i,\ \sum_{j=1}^{8} T_{ij} \leq 4,\ \forall i,\ \sum_{i=1}^{8} T_{ij} \leq 4,\ \forall j \right\}.$$

  So, each traffic matrix satisfies that every source node has a total outgoing volume of 4, and every destination node has a total incoming volume of 4.

  Your answer should either present:

  - **A linear (integer) program formulation**, or
  - **A constructive topology design** (i.e., a graph construction given as pseudocode).

  If you present a pseudocode construction, clearly explain:

  - how the topology adapts to the given traffic matrix, and
  - why the resulting topology maximizes the concurrent flow for any given traffic matrix.

  If you present a linear program, specify:

  - the decision variables,
  - the objective function, and
  - all constraints clearly.

Additionally, briefly explain the intuition behind the formulation, and how it captures the problem of finding a topology that maximizes concurrent flow for any given traffic matrix in $\mathcal{T}_{\text{hose}}$.

Finally, include the corresponding implementation using the Gurobi Python API (`https://www.gurobi.com/`). An academic license is available for free.

Report

Include a GitHub or GitLab repository link, with links to the relevant functions or line ranges. Brightspace submission should be a single PDF.

Note: Grading for this assignment will be 50% weight based on the report, and 50% weight based on the oral exam during PSO/office hours in the week of *Nov 09*.
Grace period: 3 days (72 Hours) grace period is allowed for submission. After the grace period, 25% off for every 24 hours late, rounded up.